

SONNET PODDER

PERSONAL DATA

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RESEARCH INTERESTS

Statistical Modelling • Machine Learning • Forecasting & Time-Series Analysis • Statistical Learning • Computational Statistics • Population-Level Data Analysis • Population Masking & Data Obfuscation • High-Dimensional Data Analysis • Explainable & Trustworthy AI • Privacy-Preserving Machine Learning.

EDUCATION

Northumbria University London, UK MSc Big Data and Data Science Technology (with Advanced Practice) Result: Distinction Level (Current Average: 70.5%)	January 2026 – January 2028.
Gopalganj Science and Technology University, Bangladesh. Master of Science Statistics CGPA: 3.46 out of 4	July 2022 – June 2023.
Gopalganj Science and Technology University, Bangladesh. Bachelor of Science Statistics CGPA: 3.23 out of 4	January 2017 – June 2022.

ACADEMIC PROJECTS AND RESEARCH

- 1. Statistical and Machine Learning-Based Credit Card Fraud Detection** - Developed machine learning models for highly imbalanced financial transaction data.
Used Methods - Logistic Regression, Random Forest, XGBoost, Neural Network, SMOTE, Random Under sampling, Cost-sensitive Learning, Stratified Cross Validation, Hyperparameter Tuning.
- 2. Big Data Analytics of Drug-Related Deaths** - Analysed large-scale accidental drug-related death data (2012–2024) using Python to investigate patterns and relationships in population-level data.
Used Methods - Data Cleaning, EDA, Statistical Testing, ANOVA, Correlation Analysis, Visualization
- 3. Machine Learning on Cloud: Technical Evaluation and Reflection** - Critically evaluated the fraud detection ML project from a technical and professional perspective, covering

preprocessing decisions, model selection rationale and responsible AI deployment considerations.

Used Methods - Duplicate Removal, Log Transformation, Feature Engineering, Standard Scaling, Model Comparison, False Positive/Negative Analysis, Interpretability Assessment, Fairness Evaluation, Model Monitoring.

- 4. Business Intelligence Dashboard Using Tableau** - Developed interactive dashboards from Olympic Games data to explore medal patterns across city, season, athlete age and gender.

Used Tools - Tableau, Interactive Filters, Connected Charts, Dynamic Visualisation, Dashboard Design, Visual Layout Structuring.

- 5. Machine Learning-Based Identification of Risk Factors and Determinants of Depressive Symptoms Among Elderly Bangladeshi Population** (*under preparation*) - Mohammad Kamal Hossain, Sonnet Podder. Developed a predictive modelling framework to analyse socio-demographic and health-related risk factors of geriatric depression among the elderly Bangladeshi population.

- 6. Thesis Project: Statistical Analysis of Meteorological Influence on COVID-19 Transmission in Bangladesh** - Investigated the association between COVID-19 confirmed cases and meteorological indicators including temperature and rainfall across regional epidemiological datasets in Bangladesh.

Used Methods - Spearman Correlation, Kendall Correlation, Quantile Regression, OLS Regression, Non-parametric Statistical Analysis, Python, R, SPSS.

SKILLS AND COMPETENCES

Programming Knowledge:	Python (NumPy, Pandas, Scikit-learn, XGBoost, TensorFlow, Matplotlib, Seaborn), R, SQL.
Machine Learning:	Supervised Learning, Ensemble Methods (Random Forest, XGBoost), Neural Networks, Imbalanced Learning (SMOTE, Cost-sensitive Learning), Stratified Cross-Validation, Hyperparameter Tuning, PR-AUC / ROC-AUC Evaluation
Statistical Methods:	Probability Theory, Inferential Statistics, Hypothesis Testing, Statistical Learning, Regression Modelling, Quantile Regression, Non-parametric Methods, Bayesian Statistics
Software Knowledge:	MATLAB, SPSS, STATA, JMP, Minitab, Power BI, Tableau, EndNote, Zotero.
Cloud & Tools:	Google Colab, AWS (Cloud ML Workflows), Jupyter Notebook, GitHub
Data Analysis:	Data Cleaning, Exploratory Data Analysis, Feature Engineering, Statistical Modelling, Data Visualisation, Large-

scale Data Analysis.

Research Skills:

Statistical modeling, Machine learning, predictive analytics, uncertainty quantification, risk analysis, data preprocessing, feature engineering, and large-scale data analysis using Python and R.

LANGUAGES PROFICIENCY

BENGALI: Mother Tongue

ENGLISH: Competent User [IELTS- 6.5 (L- 6.5, W-7, R- 6, S-6)]